Existential Definitions of Rings of Integers and Subfields

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Abstract

For a number field K, the established Diophantine definability of $\mathbb{Z}$ in $\mathcal{O}_K$ makes the definability of $\mathcal{O}_K$ in K equivalent to that of $\mathbb{Z}$ in K. Assuming an existential definition of $\mathbb{Z}$ in $\mathbb{Q}$, these properties are also equivalent to existential definability of the rational subfield in K. We give explicit bounds on coordinates and degree for these reductions and for descent through a finite field extension. For every finite extension L/K of characteristic-zero fields, existential definability of K in L is equivalent to definability of the relative norm image, the norm graph, or the trace graph. The norm-image implication follows from an exact finite-difference identity. In contrast, the norm and trace graphs of a proper extension are Zariski dense and have no quantifier-free definitions. Finally, a fixed finite collection of existential formulas cannot define rings of integers throughout all finite extensions of a number field. These results distinguish reductions and obstructions to transferring a formula from a counterexample to the existence of a separately chosen formula over each number field.

1 Definitions and arithmetic scope

A subset $S \subseteq F^r$ of a field F is *Diophantine in F* if it is the projection onto the first r coordinates of the common zero set of finitely many polynomials with coefficients in F. Thus there are polynomials $f_1, \dots, f_s$ such that

$$\mathbf{x} \in S \quad \Longleftrightarrow \quad \exists \mathbf{y} \in F^m : \quad f_1(\mathbf{x}, \mathbf{y}) = \cdots = f_s(\mathbf{x}, \mathbf{y}) = 0.$$

Over a field, a nonzero condition $g \neq 0$ can be expressed by $gz = 1$. Finite conjunctions, finite unions, and existential projections of Diophantine conditions are again Diophantine. For a number field, a finite system can be replaced by one equation; Lemma 2.1 gives a quantitative version. Coefficients from the specified ambient field are allowed throughout.

The *total number of coordinates* includes all distinguished input coordinates as well as all quantified coordinates. Degree is total polynomial degree. A formula with a single input and N total coordinates has $N - 1$ witnesses.

For a number field K, write $\mathcal{O}_K$ for its ring of algebraic integers. We use the established theorem

$$\mathbb{Z} \text{ is Diophantine in } \mathcal{O}_K \quad \text{for every number field } K. \tag{1}$$

The theorem is stated in [1, Corollary 1.2]; the rank-stability method and the reduction to all number fields are explained in [2, Sections 1.2 and 5]. All quantified variables in (1) range over $\mathcal{O}_K$. This distinction is essential: (1) is not itself a definition of $\mathcal{O}_K$ in K.

Where needed, we state separately the hypothesis

$$\mathbb{Z} \text{ is Diophantine in } \mathbb{Q}. \tag{H}$$

The elementary implications below do not supply a proof of (H), and their validity does not depend on a particular proposed polynomial for that hypothesis.

2 Combining equations and taking linear images

Lemma 2.1. *Over a number field F, a system of $s \geq 1$ polynomial equations of degree at most D can be replaced, in the same coordinates, by one equation of degree at most $\lceil \sqrt{s} \rceil D$. If F has a real embedding and $s > 1$, the bound $2D$ also holds. For $s = 1$, the original equation suffices.*

Proof. Put $e = \lceil \sqrt{s} \rceil$. Choose a finite prime $\mathfrak{p}$ of F, an element π of valuation one there, and a monic polynomial g of degree e with integral coefficients and irreducible reduction over the residue field. Let

$$N_g(Z_0, \ldots, Z_{e-1}) = \det\bigl(m_{Z_0 + Z_1 u + \cdots + Z_{e-1} u^{e-1}}\bigr)$$

be the multiplication determinant in $F[u] = F[U]/(g)$. For any nonzero block $\mathbf{z} \in F^e$, reduction after scaling by its least coordinate valuation gives

$$v_{\mathfrak{p}}(N_g(\mathbf{z})) = e \min_i v_{\mathfrak{p}}(z_i).$$

Indeed, the reduced multiplication matrix multiplies by a nonzero element of the residue extension field, so its determinant is a unit. Therefore the degree-e form

$$A_e(\mathbf{z}_0, \ldots, \mathbf{z}_{e-1}) = \sum_{j=0}^{e-1} \pi^j N_g(\mathbf{z}_j)$$

vanishes only when all e^2 entries are zero: its nonzero summands have distinct valuation classes modulo e, so a unique term has least valuation. Pad the original residuals to length e^2 and substitute them into A_e.

If F has a real embedding, a sum of squares vanishes in F only when each summand vanishes. The sum of the squares of the residuals gives the second bound. □

The following elementary observation avoids an unnecessary coordinate in several constructions.

Lemma 2.2. *For $1 \leq i \leq n$, let $S_i \subseteq F$ be defined by a polynomial $F_i(T, \mathbf{Y}_i)$ with N_i total coordinates and degree at most D. Fix $b_1, \ldots, b_n \in F$, with $b_n \neq 0$. The set*

$$b_1 S_1 + \cdots + b_n S_n$$

has a definition with $\sum_i N_i$ total coordinates, n equations, and degree at most D.

Proof. For the distinguished input x, retain $z_1, \ldots, z_{n-1}$ and put

$$z_n = \frac{x - \sum_{i<n} b_i z_i}{b_n}.$$

Impose $F_i(z_i, \mathbf{Y}_i) = 0$ for every i, using disjoint witness blocks. There are

$$1 + (n-1) + \sum_i (N_i - 1) = \sum_i N_i$$

coordinates. Substitution of a linear expression does not increase total degree. The equivalence with membership in the displayed linear image is immediate in both directions. □

3 Rational integers, algebraic integers, and rational inputs

Theorem 3.1. *For every number field K, the following assertions are equivalent:*

1. *$\mathcal{O}_K$ is Diophantine in K;*
2. *$\mathbb{Z}$ is Diophantine in K.*

Under hypothesis (H), they are also equivalent to:

3. *the subfield $\mathbb{Q}$ is Diophantine in K.*

Proof. Assume (1). Take a definition supplied by (1) and replace the requirement that each input or witness lie in $\mathcal{O}_K$ by its assumed existential definition in K. All quantifiers remain existential. Restriction of every original variable to its original ring makes the new formula define exactly $\mathbb{Z}$.

Assume (2). Choose an integral basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}_K$ over $\mathbb{Z}$. Then

$$x \in \mathcal{O}_K \quad \Longleftrightarrow \quad \exists z_1, \dots, z_n \in \mathbb{Z} : \quad x = \sum_{i=1}^{n} \omega_i z_i.$$

Replacing the integer membership conditions by their definitions inside K proves (1).

Still under (2), rationality is expressed by

$$q \in \mathbb{Q} \quad \Longleftrightarrow \quad \exists a, b \in \mathbb{Z}\ \exists c \in K : qb = a, \qquad bc = 1.$$

Thus (2) implies (3), without using (H).

Finally assume (3) and (H). In the defining formula over $\mathbb{Q}$, restrict its input and every witness to the Diophantine subfield $\mathbb{Q} \subseteq K$. The resulting existential formula over K is equivalent to the original one over $\mathbb{Q}$ and defines $\mathbb{Z}$. □

Proposition 3.2. *Let $n = [K : \mathbb{Q}]$. If $\mathbb{Z}$ has a definition in K with N total coordinates and degree D, then $\mathcal{O}_K$ has a definition with nN total coordinates and n equations of degree at most D. It has a single-equation definition in those coordinates of degree at most $\lceil \sqrt{n} \rceil D$, or at most $2D$ if K has a real embedding and $n > 1$.*

Proof. Apply Lemma 2.2 to the integral-basis identity in the proof of Theorem 3.1, then apply Lemma 2.1. □

Proposition 3.3. *Suppose that a polynomial of degree D and N total coordinates defines $\mathbb{Z}$ in $\mathbb{Q}$, and a polynomial of degree E and R total coordinates defines $\mathbb{Q}$ in a number field K. Then $\mathbb{Z}$ has a definition inside K with NR total coordinates, $N + 1$ equations, and degree at most $\max(D, E)$. The ring $\mathcal{O}_K$ has a definition with nNR total coordinates, $n(N+1)$ equations, and the same degree bound, where $n = [K : \mathbb{Q}]$.*

Proof. Write the first polynomial as $P(T_0, \dots, T_{N-1})$ and the second as $Q(T, \mathbf{V})$, with $R - 1$ witnesses in $\mathbf{V}$. Impose

$$P(T_0, \dots, T_{N-1}) = 0, \qquad Q(T_i, \mathbf{V}_i) = 0 \quad (0 \le i < N).$$

The N original coordinates and the $N(R-1)$ new ones total NR. The argument in Theorem 3.1 proves correctness. For $\mathcal{O}_K$, use n copies of this system in the integral-basis construction and eliminate the last integer coordinate as in Lemma 2.2. This yields the stated count without first aggregating either system. □

Remark 3.4. In the first part of Proposition 3.3, restricting only the witnesses is insufficient in general: a polynomial with rational coefficients and rational values for the witnesses can still have nonrational algebraic roots in its input. In particular, the polynomial defining $\mathbb{Z}$ over $\mathbb{Q}$ cannot simply be evaluated with unrestricted K-valued witnesses.

4 Relative norms and traces characterize the missing subfield

Throughout this section, L/K is a finite extension of characteristic-zero fields of degree n. The norm and trace are the usual determinant and trace of multiplication on the n-dimensional K-vector space L. Define the following subsets of powers of the *ambient field* L:

$$T_{L/K} = \{\mathrm{N}_{L/K}(a) : a \in L\} \subseteq K \subseteq L,$$
$$G_N = \{(a, b) \in L^2 : b = \mathrm{N}_{L/K}(a)\},$$
$$G_T = \{(a, b) \in L^2 : b = \mathrm{Tr}_{L/K}(a)\}.$$

In particular, definability here does not silently require the coordinates of a in a K-basis to belong to K.

Lemma 4.1. *For every integer $n \geq 1$, one has the polynomial identity*

$$\sum_{j=0}^{n-1} (-1)^{n-1-j} \binom{n-1}{j} (T + j)^n = n!T + \frac{n!(n-1)}{2}. \tag{2}$$

If $S \subseteq K$ contains every nth power of an element of K, then

$$K = \sum_{j=0}^{n-1} (-1)^{n-1-j} \binom{n-1}{j} S. \tag{3}$$

Proof. The left side of (2) is the $(n-1)$st forward difference of T^n. It is linear with coefficient $n!$. Its constant term equals $(n-1)!S(n, n-1)$, where $S(n, n-1)$ counts partitions of an n-element set into $n-1$ nonempty parts. Exactly one pair is joined, so $S(n, n-1) = \binom{n}{2}$, giving the stated constant. This also follows directly by inclusion and exclusion applied to surjections from an n-element set to an $(n-1)$-element set. For $n = 1$, the identity is $T = T$.

Every element on the right of (3) belongs to K. Conversely, given $q \in K$, choose $T = q/n! - (n-1)/2 \in K$ in (2). Each $(T + j)^n$ belongs to S, and their indicated linear combination is q. □

Theorem 4.2. *The following assertions are equivalent:*

1. *K is Diophantine in L;*
2. *the norm image $T_{L/K}$ is Diophantine in L;*
3. *the norm graph G_N is Diophantine in L;*
4. *the trace graph G_T is Diophantine in L.*

If $T_{L/K}$ is defined by a polynomial with M total coordinates and degree D, then K has a definition with nM total coordinates and n equations of degree at most D.

Proof. Assume (1), and choose a K-basis $b_1, \ldots, b_n$ of L. For $a \in L$, write $a = \sum_i a_i b_i$ with all $a_i \in K$. The entries of its multiplication matrix are fixed K-linear forms in the a_i. Its determinant and trace are therefore polynomials over K in these coordinates. Restricting each a_i to the given Diophantine subfield K makes the graph of each map Diophantine in L. Projection of the norm graph proves (2) as well as (3) and (4).

Assertion (3) implies (2) by projection. If (2) holds, then $T_{L/K} \subseteq K$ and $\mathrm{N}_{L/K}(q) = q^n$ for every $q \in K$. Lemma 4.1 expresses all of K as a fixed linear image of n copies of $T_{L/K}$, proving (1). Lemma 2.2 proves the quantitative assertion, since the last coefficient in (3) is one.

Finally the trace is onto K: for $q \in K$, $\mathrm{Tr}_{L/K}(q/n) = q$. Projection of G_T therefore proves that (4) implies (1). □

Corollary 4.3. *If L/K is quadratic, existential definability in L of any of the four sets in Theorem 4.2 is also equivalent to existential definability of the graph of the nontrivial K-automorphism σ of L. Moreover,*

$$x \in K \iff \sigma(x) = x \iff \mathrm{Tr}_{L/K}(x) = 2x \iff \mathrm{N}_{L/K}(x) = x^2.$$

Proof. A definition of the automorphism graph defines its fixed field. Conversely, if K is Diophantine in L, write $x = a + b\theta$ with $a, b \in K$ and use $\sigma(x) = a + b\sigma(\theta)$. The trace equivalence follows from $\mathrm{Tr}(x) = x + \sigma(x)$. For the norm, $\mathrm{N}(x) = x\sigma(x) = x^2$ implies either $x = 0$ or $\sigma(x) = x$; zero already belongs to K. □

For example, when $n = 2$, the norm-image identity is simply

$$K = T_{L/K} - T_{L/K}.$$

Indeed, for $q \in K$, the identity $((q+1)/2)^2 - ((q-1)/2)^2 = q$ exhibits the difference of two norms of elements of K itself.

4.1 Why an ordinary norm form does not solve this problem

A norm form with *K-valued coordinates* is an ordinary polynomial over K. The map on a *single L-valued input* addressed in Theorem 4.2 is a different object. The distinction has the following exact algebraic manifestation.

Proposition 4.4. *If L/K is proper, the graphs G_N and G_T are Zariski dense in $\mathbb{A}^2_L$. Neither has a quantifier-free definition in the language of rings with coefficients in L. In particular neither is the graph of a polynomial function in one variable over L.*

Proof. Choose $\theta \in L \setminus K$. The set K^2 is Zariski dense in $\mathbb{A}^2_L$, because a polynomial vanishing on K^2 is zero by applying the one-variable root bound successively.

Consider first the trace graph. Its points include the images of K^2 under the L-linear map

$$(a, b) \longmapsto (a + b\theta,\ na + b\,\mathrm{Tr}_{L/K}(\theta)).$$

Its determinant is $\mathrm{Tr}_{L/K}(\theta) - n\theta \neq 0$: otherwise $\theta \in K$. Hence this map is an automorphism of $\mathbb{A}^2_L$, proving density of G_T.

For the norm, restrict the polynomial norm form to the two coordinates a, b corresponding to $a + b\theta$ and consider

$$(a, b) \longmapsto (a + b\theta,\ \mathrm{N}_{L/K}(a + b\theta)).$$

At $(1, 0)$ its Jacobian determinant is again $\mathrm{Tr}_{L/K}(\theta) - n\theta \neq 0$, since the linear term in $\mathrm{N}(1 + b\theta)$ is $b\,\mathrm{Tr}(\theta)$. The map is dominant. Its values on K^2 belong to G_N, so G_N is dense as well.

A quantifier-free formula defines a constructible set. A dense constructible subset of $\mathbb{A}^2_L$ contains a nonempty Zariski open subset. Since L is infinite, such an open subset has infinitely many points in at least one vertical line; for instance, choose a first coordinate at which a nonzero defining polynomial of a principal open subset remains nonzero. A graph of a function has exactly one point in each vertical line. This contradiction rules out a quantifier-free definition. A polynomial graph is a proper closed subset and is excluded in particular. □

5 Descent and the structure of a possible counterexample

Theorem 5.1. *Let L/K be a finite extension of number fields of degree d. If $\mathcal{O}_L$ has a definition in L by a polynomial with N total coordinates and degree D, then $\mathcal{O}_K$ has a definition in K with $1 + d(N-1)$ total coordinates, d equations, and degree at most D. A single equation in those coordinates can be chosen of degree at most $\lceil\sqrt{d}\rceil D$; if K has a real embedding and $d > 1$, the bound $2D$ also applies.*

Proof. Fix a K-basis $b_1, \dots, b_d$ of L and write the defining polynomial as $P(T, Y_1, \dots, Y_{N-1})$. Keep the input $x \in K$ and substitute

$$Y_j = \sum_{i=1}^{d} b_i y_{ji}, \qquad y_{ji} \in K.$$

Expanding in the basis gives a unique expression

$$P\left(x, \sum_i b_i y_{1i}, \dots, \sum_i b_i y_{N-1,i}\right) = \sum_{i=1}^{d} b_i P_i(x, (y_{ji})),$$

where every P_i belongs to $K[x, (y_{ji})]$ and has degree at most D. Vanishing is equivalent to all $P_i = 0$ by linear independence. The substitutions range over all original witnesses in L, so the input projection is $\mathcal{O}_L \cap K = \mathcal{O}_K$. The coordinate count and aggregation bounds follow immediately. □

Corollary 5.2. *If some number field K has a ring of integers that is not Diophantine in K, then every finite extension of K has the same property. In particular, there exists a totally imaginary number field, Galois over $\mathbb{Q}$, with that property.*

Proof. The first assertion is the contrapositive of Theorem 5.1. Take a normal closure of $K/\mathbb{Q}$ and adjoin i. The resulting field is finite and Galois over $\mathbb{Q}$, contains K, and has no real embedding. □

This is a reduction of a possible counterexample, not the construction of one. In particular, a polynomial that fails after base extension does not establish the hypothesis of Corollary 5.2.

6 A finite collection of unchanged formulas is insufficient

An existential formula in this section may use finite Boolean combinations of polynomial equalities, so nonzero conditions and finite alternatives are allowed. Coefficients are fixed in a number field k. Its interpretation in an extension L/k uses the same formula and the same coefficients, with witnesses in L.

Lemma 6.1. *Let $\Phi(T)$ be such a formula and suppose $\Phi(k)$ is infinite. There are an integer $d \geq 1$ and a nonzero $h \in k[T]$ such that, for every $a \in k$ with $h(a) \neq 0$, the formula $\Phi(a)$ holds in some extension E_a/k of degree at most d.*

Proof. The constructible set of witness tuples has a constructible image in the input line by Chevalley's theorem [4]. An infinite image is dense in that line, so one conjunction in the formula has nonempty generic fiber over $k(T)$. Choose a closed point of this generic fiber, with residue field E finite over $k(T)$. Write $E = k(T)[\alpha]$, with monic minimal polynomial $G(T, Z)$ of degree d in Z. Express the witness coordinates and the inverses of all required nonzero expressions as polynomials in α with coefficients in $k(T)$.

After inverting a nonzero $h \in k[T]$, all these expressions and the coefficients of G belong to $R = k[T, 1/h]$. The algebra $C = R[Z]/(G)$ is free of rank d and embeds in E. All equations and inverse identities therefore hold in C. For $h(a) \neq 0$, specialize to $C_a = k[Z]/(G(a, Z))$ and pass to any residue field at a maximal ideal. Its degree over k is at most d, and all equations and nonzero conditions survive. □

Theorem 6.2. *Let $\Phi_1, \dots, \Phi_r$ be a finite nonempty list of existential formulas over a number field k. Suppose $\Phi_i(k)$ is infinite for every i. There exists $B \geq 1$ such that, for all but finitely many $q \in \mathbb{Q} \setminus \mathbb{Z}$, there is a finite Galois extension L_q/k of degree at most B in which every $\Phi_i(q)$ holds. Thus none of the formulas defines $\mathcal{O}_{L_q}$ in L_q. One may further require L_q to have no real embedding, at the cost of at most doubling B.*

Proof. Apply Lemma 6.1 to each formula, obtaining d_i and h_i. Exclude the finite union of the rational zero sets of the h_i. For every remaining rational noninteger q, choose the witnessing fields $E_{i,q}$. The normal closure of $E_{i,q}/k$ has degree at most $d_i!$, by its permutation action on the conjugates of a primitive element. Take L_q to be the compositum of these normal closures. Then L_q/k is Galois of degree at most

$$B = \prod_{i=1}^{r} d_i!.$$

Each witness tuple remains a witness in L_q: polynomial equalities and nonzero conditions are preserved by field extensions. A rational algebraic integer is an integer, so $q \notin \mathcal{O}_{L_q}$. Adjoining i gives the last assertion. □

Corollary 6.3. *A fixed finite collection of formulas over k, each correct for $\mathcal{O}_k$ in k, cannot supply a definition of the ring of integers in every finite extension by choosing one formula from that collection for each extension.*

Proof. Each formula has infinitely many inputs over k. The extension in Theorem 6.2 makes every formula in the collection incorrect simultaneously. □

For a single nonzero polynomial $P(T, \mathbf{Y})$ that depends on a witness coordinate Y_j, the integer d in Lemma 6.1 may be chosen as $\deg_{Y_j} P$. Indeed, specialize all other witnesses to constants in k so that the leading coefficient in Y_j remains a nonzero polynomial of T. Outside its finite zero set, adjoining a root gives the asserted extension. Consequently the factorial bound in Theorem 6.2 can use the smallest positive individual witness degree of each polynomial.

7 What rank stability supplies inside a field

A positive-rank elliptic curve whose rank does not grow in an extension supplies a useful Diophantine subset of the smaller field. It is necessary to distinguish this conclusion from existential definability of the entire smaller field.

Proposition 7.1. *Let K be a number field and let $E/\mathbb{Q}$ be an elliptic curve such that*

$$0 < \operatorname{rank} E(\mathbb{Q}) = \operatorname{rank} E(K).$$

There is an integer $r \geq 1$ for which the set

$$T = \{x(P) : P \in rE(K),\ P \neq O\}$$

is an infinite Diophantine subset of K contained in $\mathbb{Q}$.

Proof. The Mordell–Weil groups are finitely generated and have the same rank, so $E(K)/E(\mathbb{Q})$ is finite. Choose r annihilating this quotient. Then $rE(K) \subseteq E(\mathbb{Q})$, and $rE(K)$ is infinite. The graph of multiplication by the fixed integer r is algebraic. On a finite collection of affine charts it is given by polynomial equations and nonzero conditions. Projecting its nonidentity image onto the abscissa coordinate gives a Diophantine set over K. The abscissa map has finite fibers, so this set is infinite; all its elements are rational because its points lie in $E(\mathbb{Q})$. □

The known arithmetic transfer theorem under the same rank hypothesis defines the smaller *ring of integers* inside the larger ring of integers; the more general abelian-variety version is [2, Theorem 4.2]. For fields themselves, [3, Theorem 4] gives a definition of the smaller subfield using one universal quantifier. Remark 6.3 of [3] expressly distinguishes that conclusion from an existential definition in the whole field. Those results therefore do not remove the rational-input restriction in Proposition 3.3.

In general, taking multiplicative combinations of the abscissas in Proposition 7.1 does not yield every rational number. The following obstruction gives an explicit example.

Lemma 7.2. *Let*

$$E : y^2 = x^3 + Ax + B$$

be a nonsingular elliptic curve over $\mathbb{Q}$. Suppose that p is an odd prime, A and B are p-integral, and the reduction of B is a nonsquare in $\mathbb{F}_p^\times$. Every affine rational point of E has nonzero abscissa x and even valuation $v_p(x)$. The multiplicative subgroup of $\mathbb{Q}^\times$ generated by all these abscissas consists entirely of numbers of even p-adic valuation.

Proof. If $x = 0$, then $y^2 = B$, contrary to the nonsquare residue of B. If $v_p(x) > 0$, the equation has a unit right-hand side congruent to B modulo p, giving the same contradiction. If $v_p(x) < 0$, then

$$3v_p(x) < v_p(Ax), \qquad 3v_p(x) < v_p(B).$$

The term x^3 has the unique least valuation on the right, so $2v_p(y) = 3v_p(x)$. This forces $v_p(x)$ to be even. The remaining case has valuation zero. Valuations add under multiplication and change sign under inversion, proving the last assertion. □

Proposition 7.3. *The curve $E : y^2 = x^3 - 2$ has a rational point of infinite order, but 5 is not a ratio of two rational abscissas of E, or any finite product of integral powers of its rational abscissas.*

Proof. The point $(3, 5)$ lies on E. The curve has good reduction at 5 and 7, since its discriminant is -1728. Direct counting gives

$$\#E(\mathbb{F}_5) = 6, \#E(\mathbb{F}_7) = 7.$$

For completeness, modulo 5 the numbers of ordinates above $x = 0, 1, 2, 3, 4$ are $0, 2, 2, 1, 0$. Modulo 7 the corresponding numbers are $0, 0, 0, 2, 0, 2, 2$. Add the identity in each case.

Prime-to-p rational torsion injects under good reduction at p. One way to see the needed injectivity is to use the formal parameter $z = -x/y$ in the reduction kernel: if $p \nmid m$, then $[m]z = mz + O(z^2)$ has the same finite valuation as z and cannot vanish. Thus a prime divisor $\ell \neq 5, 7$ of a rational torsion order would divide both 6 and 7, which is impossible. The 5-primary torsion is excluded by reduction at 7, and the 7-primary torsion by reduction at 5. Hence $E(\mathbb{Q})$ has trivial torsion, and $(3, 5) \neq O$ has infinite order.

At $p = 5$, the constant coefficient is $B = -2 \equiv 3$, a nonsquare. Lemma 7.2 shows that every multiplicative combination in the statement has even 5-adic valuation, whereas $v_5(5) = 1$. □

The obstruction in Proposition 7.3 concerns multiplicative combinations. It does not apply to sums of such values and does not disprove a more elaborate existential definition. In particular, it supplies neither a counterexample to Theorem 3.1's equivalent properties nor a positive result for a new number field.

8 Consequences for constructions over number fields

The distinction between two questions is exact. Theorems 5.1 and 3.1 concern the existence of a formula chosen for a specified field. Theorem 6.2 concerns reusing a fixed finite collection of formulas throughout extensions. The second question has a negative answer even if the first has a positive answer for every number field.

Under (H), a proof that $\mathcal{O}_K$ is Diophantine in K can therefore proceed by constructing any one of the following: a definition of the rational subfield; a definition of the absolute norm image as a subset of K; or a definition of the absolute trace graph with witnesses in K. Theorem 4.2 shows that these are equivalent arithmetic tasks, rather than independent shortcuts obtained by

writing a formal norm or trace symbol. Alternatively, a direct definition of any subset $S \subseteq \mathcal{O}_K$ containing $\mathbb{Z}$ is sufficient: the integral-basis construction gives

$$\mathcal{O}_K = \sum_{i=1}^{n} \omega_i S,$$

because the right side is contained in $\mathcal{O}_K$ and contains its full integral-basis expansion. This last sufficient condition does not assume (H).

No theorem above gives an example of a number field whose ring of integers is not Diophantine in its fraction field, or a proof that all number fields have the property. The remaining step is an arithmetic definition inside the field itself, with all witnesses in the field and with soundness against nonintegral inputs.

References

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